Quality Control (QC) of Sequencing Data

Shahiid Kiyaga

MRC/UVRI and LSHTM Uganda Research Unit

Objectives

Learning Objectives:

- Understand the FASTQ file format and Phred quality scores.
- Perform QC on sequencing reads using FastQC, MultiQC, and fastp.
- Install and manage bioinformatics tools with Conda environments.
- Apply QC best practices for naming files and ensuring high-quality data for analysis.

Why Sequence Quality Control Matters?

- **Reliable Results:**

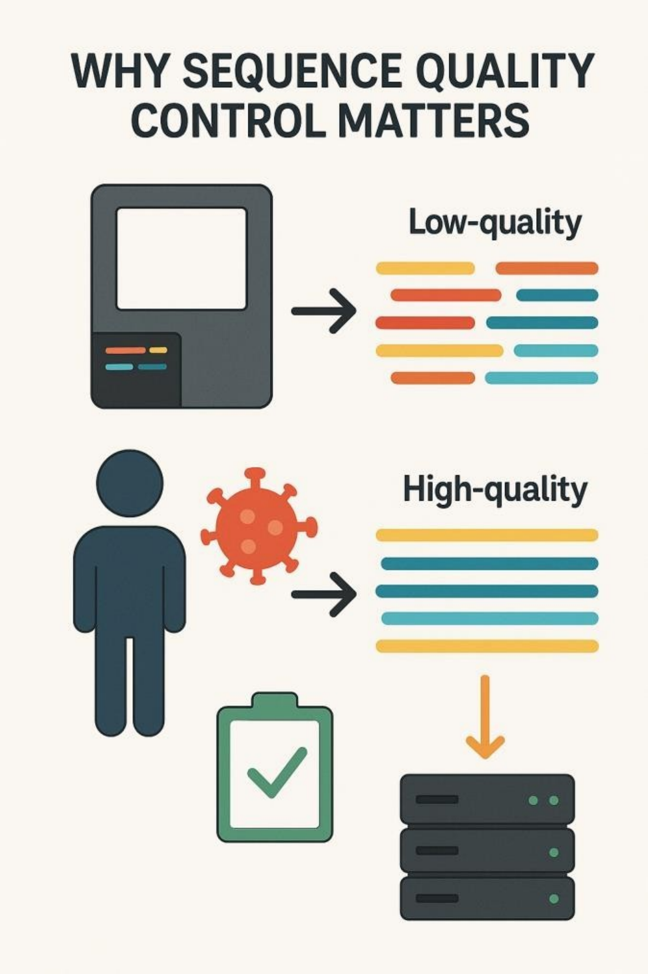
- Downstream analyses (assemblies, variant calling) depend input read quality.

- Low-quality bases can lead to incorrect genome assemblies or false variant calls, undermining outbreak investigations or AMR surveillance efforts.

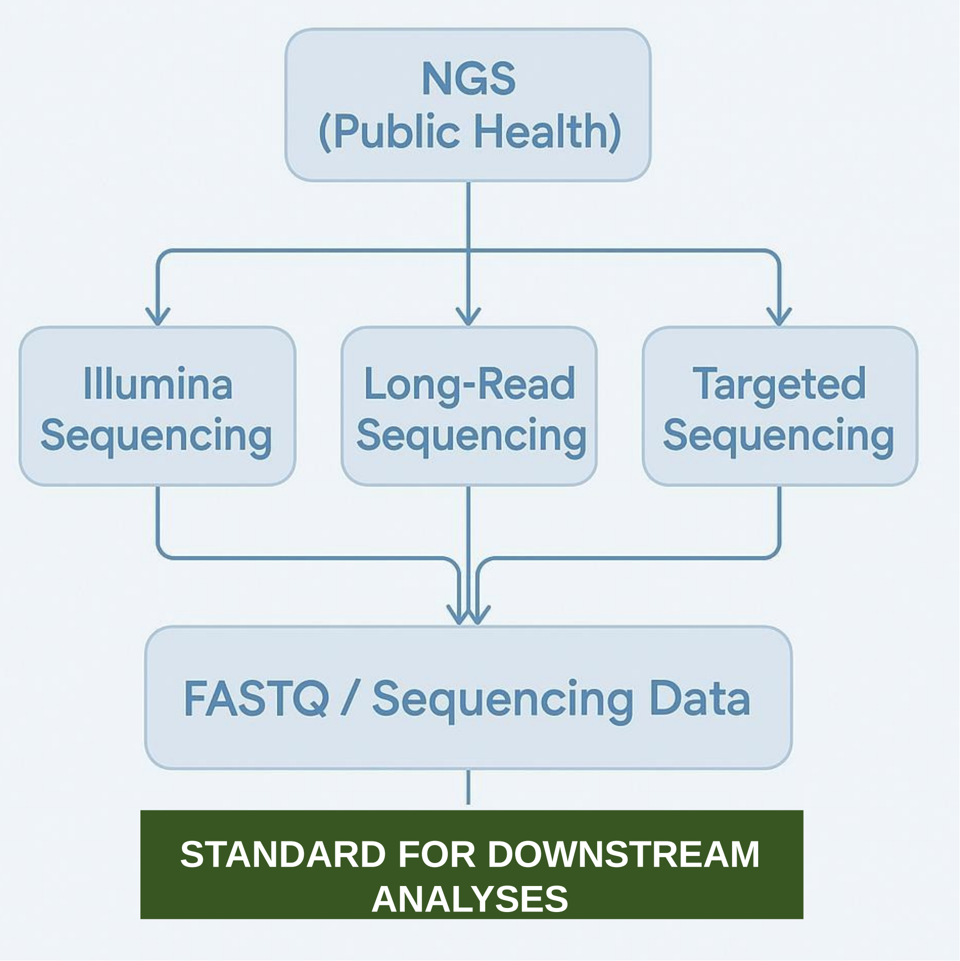
- **Efficiency:**

→ **QC saves resources** by flagging bad libraries or contamination early. It prevents wasted compute on **garbage data**

→ guides whether re-sequencing is needed.



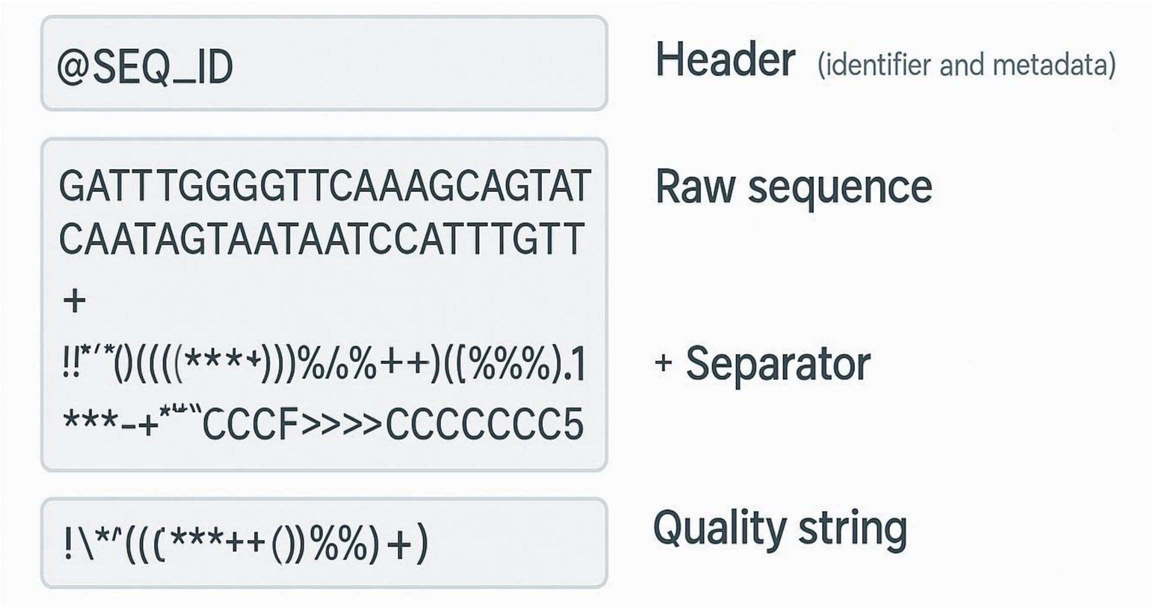
Sequencing Data in Pathogen Genomics



- **Illumina Sequencing**: Short-read, high accuracy; widely used in pathogen genomics.
- **Long-Read Sequencing**: (e.g., Oxford Nanopore, PacBio) captures structural variants, used in genome assembly and finishing.
- **Targeted Sequencing**: Focuses on specific genes or regions (e.g., AMR genes, 16S); faster and cost-effective.
- **All sequencing technologies** output reads in **FASTQ format**, which contains base calls and quality scores.
- **FASTQ files** are the universal input for downstream QC, assembly, alignment, and variant analysis pipelines.
- **Standardization to FASTQ** ensures compatibility across tools, workflows, and public health databases.

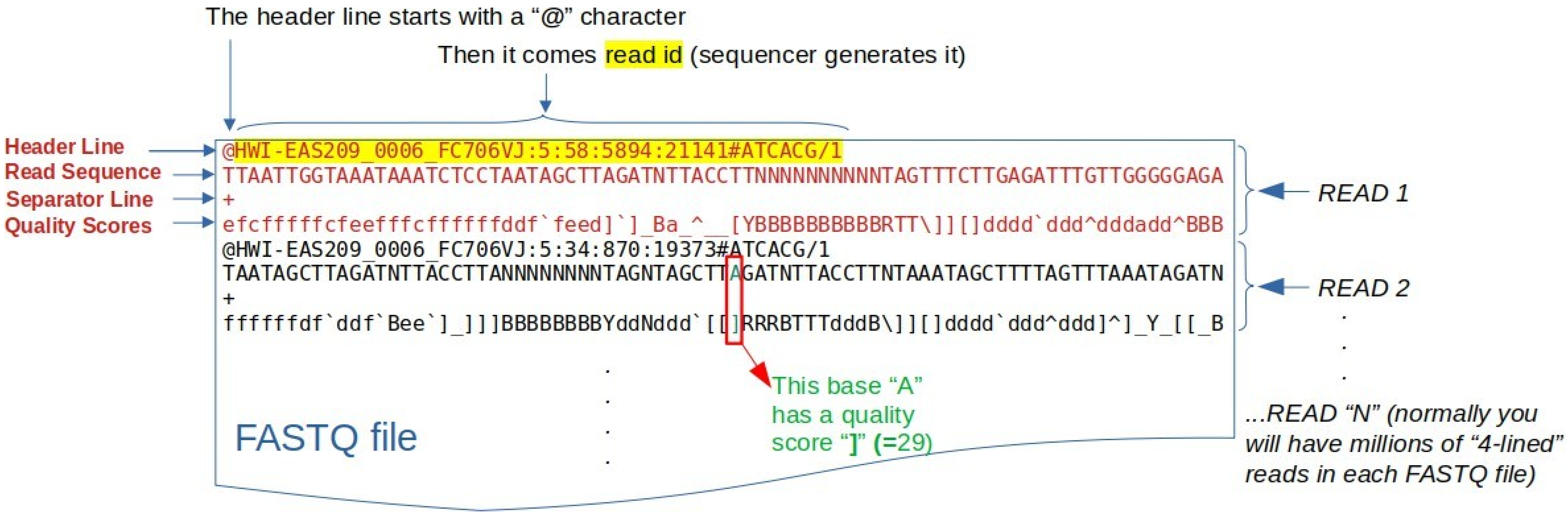
FASTQ Format Basics

Purpose: FASTQ allows both sequence and quality to be stored compactly and used in QC, alignment, and variant calling workflows.



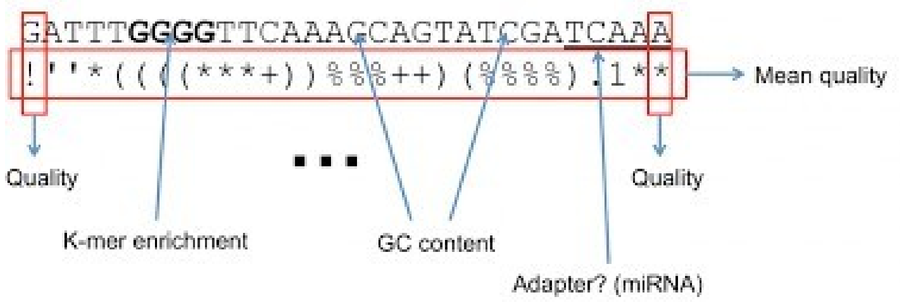
- **@ Header line**: Starts with @, contains a unique identifier and optional metadata about the sequencing read.
- **Raw sequence line**: Nucleotide bases (A, T, G, C, N); represents the DNA or RNA fragment.
- **+ Separator line:** Begins with +; optionally repeats the identifier but often left blank.
- **Quality string line:** Encodes per-base quality scores using ASCII characters.
- **Phred+33 encoding (Illumina 1.9+)**: Each ASCII character maps to a quality score (e.g., ! = Q0, ? = Q30, ~ = Q93).
- **Length consistency**: Sequence and quality lines must be the same length—one quality score per base.

FASTQ Format Example



Phred Quality Scores and Accuracy

$$Q = -10 \log_{10} P \longrightarrow P = 10^{\frac{-Q}{10}}$$



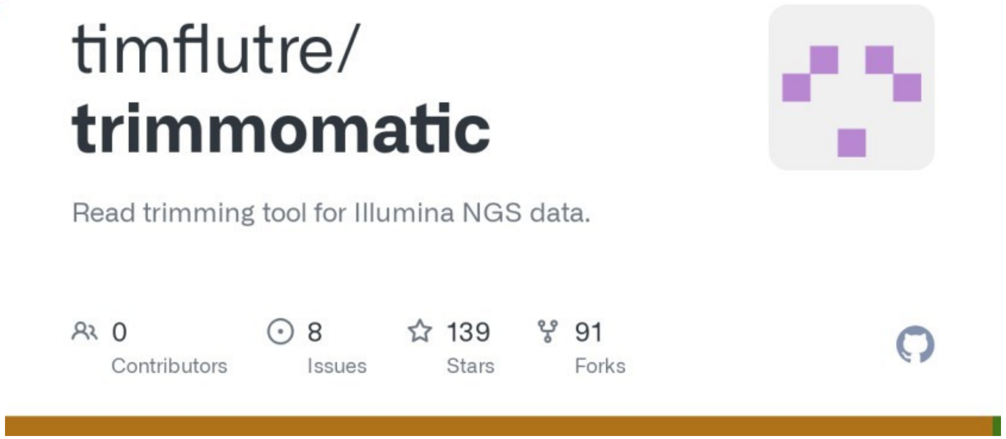
Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

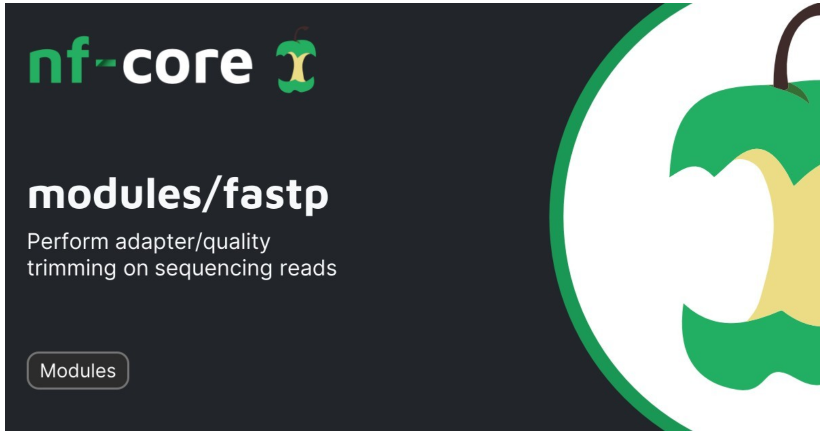
- Sequencing instruments assign each base a **Phred quality score (Q)** reflecting confidence in the base call.
- The score is logarithmic: $Q = -10\log_{10}(P_error)$This means a **10-point increase** in Q is a **10-fold reduction** in error probability.
- Common benchmarks: Q20 ≈ 1% error (99% accuracy), Q30 ≈ 0.1% error (99.9% accuracy), Q40 ≈ 0.01% error (99.99% accuracy)
- In practice, **aim for ≥Q30** for the majority of bases (often referred to as *"Q30"* as a quality metric).

Tools for QC of NGS Data (Ilumina Short Reads)

FastQC

MultiQC





Tools for QC of NGS Data (Ilumina Short Reads)

FastQC

- Diagnostic tool for raw FASTQ reads.
- Produces per-sample HTML reports with modules like:
  - Per-base quality
  - GC content
  - Sequence length distribution
  - Overrepresented sequences
- Flags issues (**PASS**/**WARN**/**FAIL**) but must be interpreted in context.
- **Used to assess, not modify, read data !!**

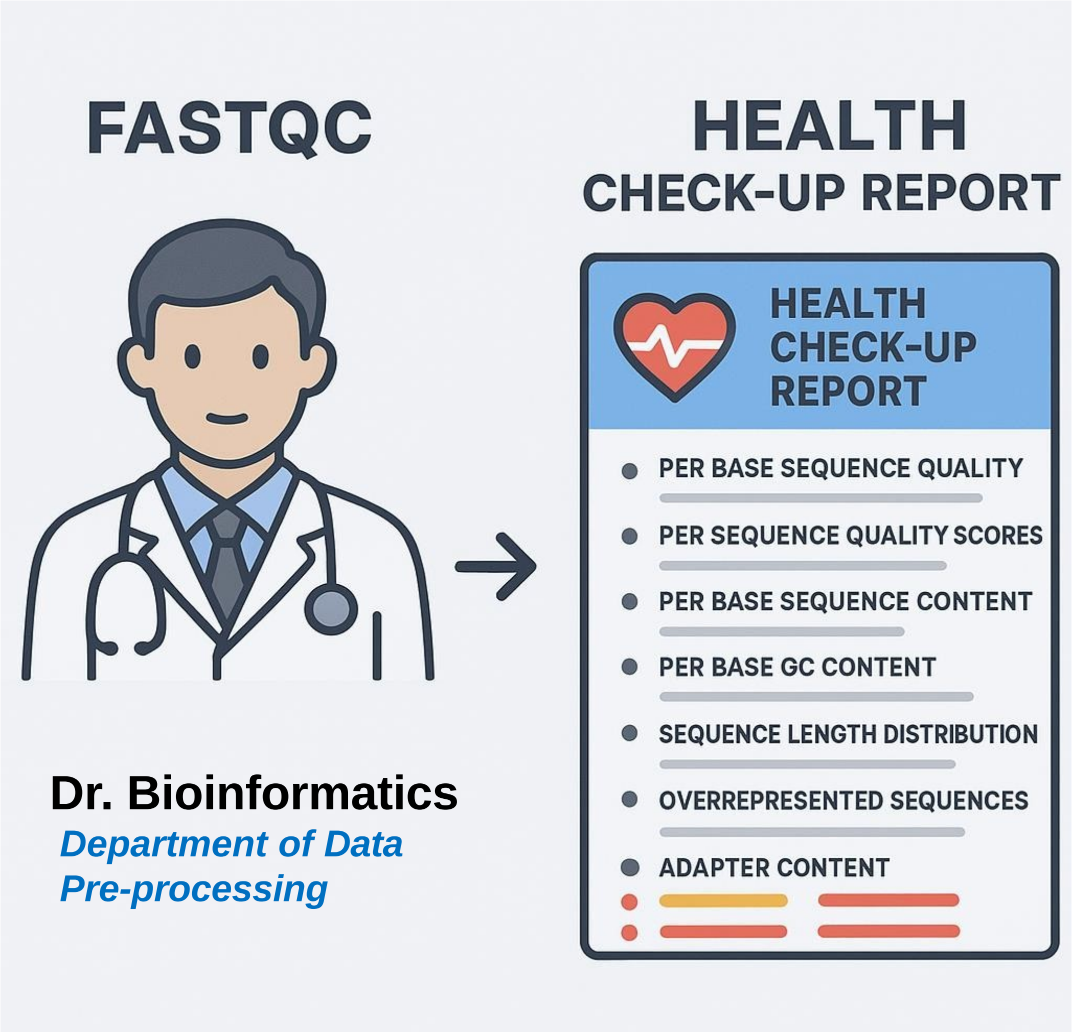
FastQC Modules vs Medical Check-Up

- **Per Base Sequence Quality** → Blood Pressure Measures how healthy each position in the read is. **Sudden drops = a problem.**

- **Per Sequence Quality Scores** → Overall Health Score Are most reads healthy or are many poor quality?

- **Per Base Sequence Content** → Nutrition Panel Too much of one base (like too much sugar) signals a problem—maybe adapter contamination or bias.

- **Per Base GC Content** → Metabolic Balance Deviation from **expected GC%** hints at contamination or biased library prep.

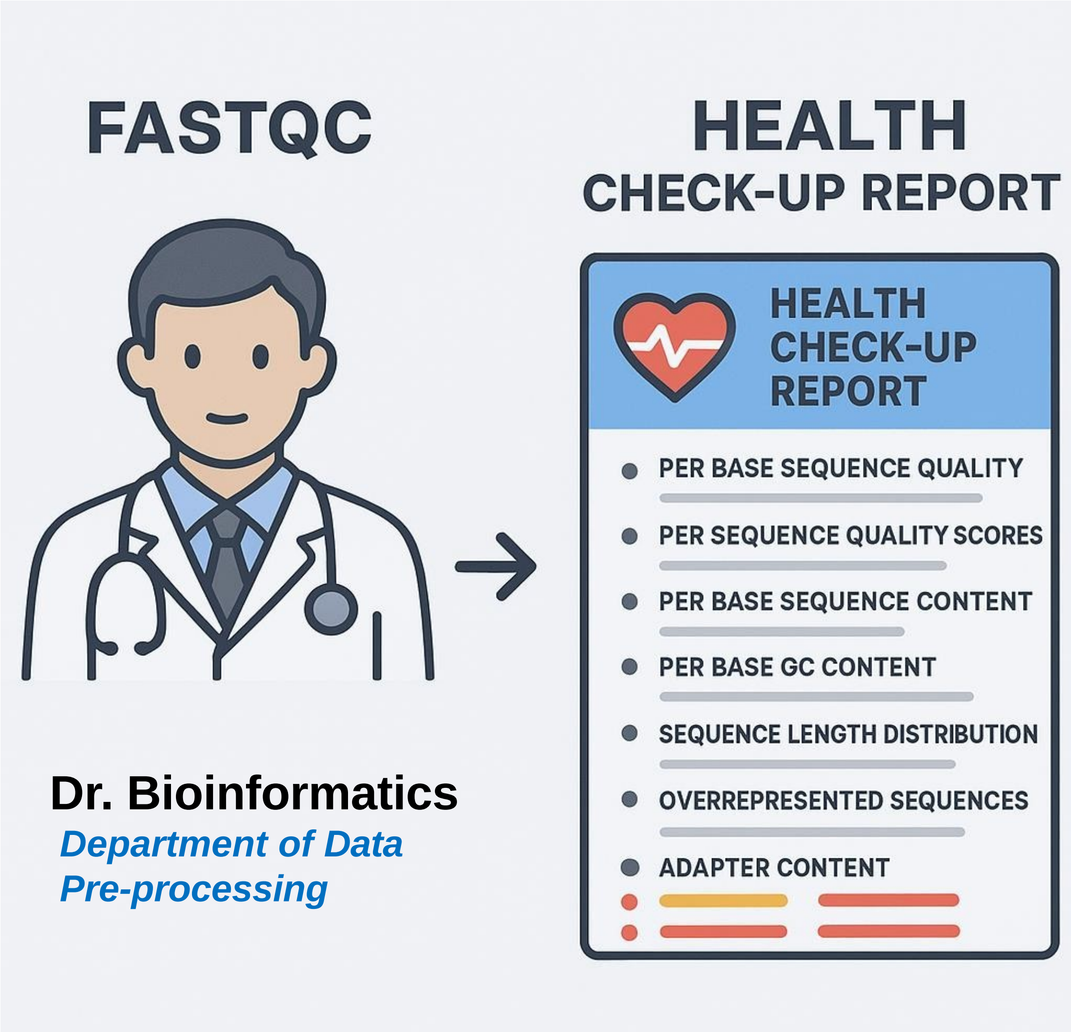


FastQC Modules vs Medical Check-Up

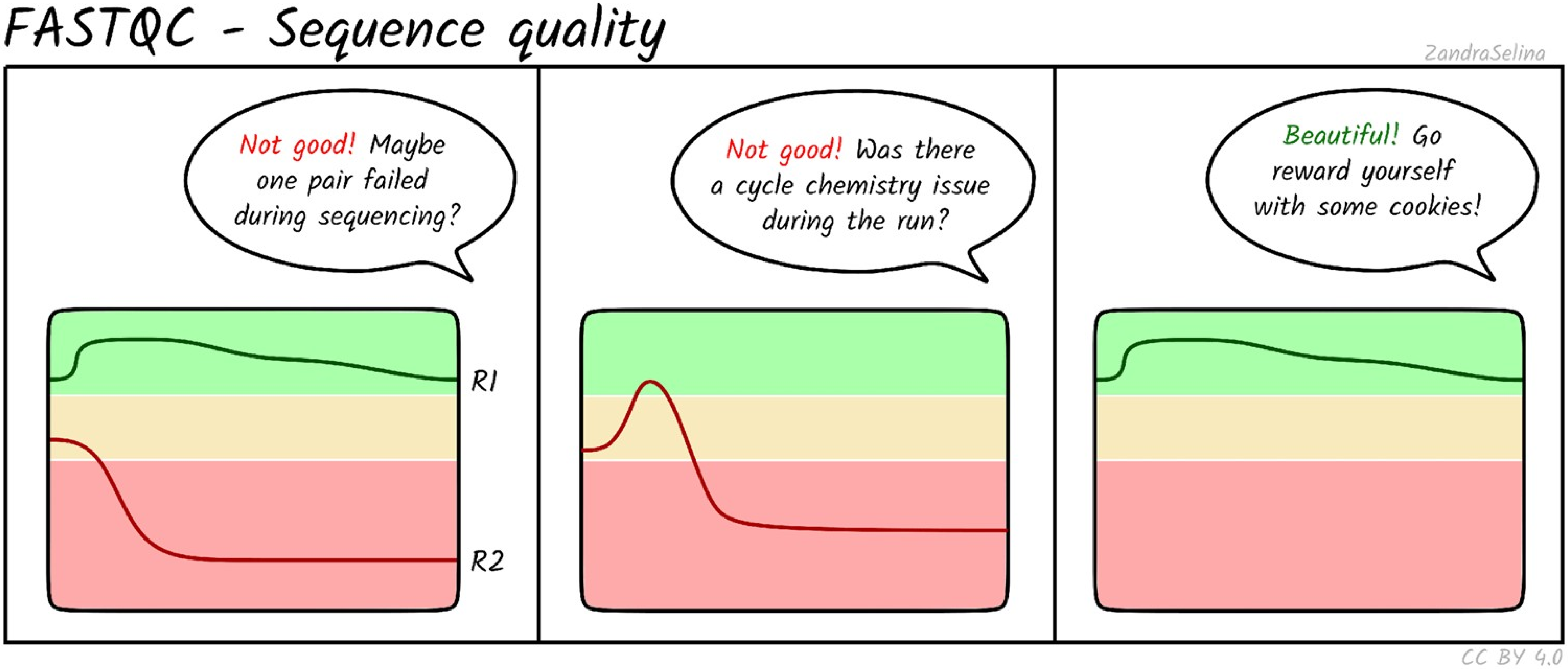
- **Sequence Length Distribution** → Height Chart
  Are your reads the expected length? Shorter or mixed sizes may indicate trimming or fragmentation issues.

- **Overrepresented Sequences** → **Red Flag Symptoms**
  Sequences that occur too often might be contamination, adapters, or barcodes.

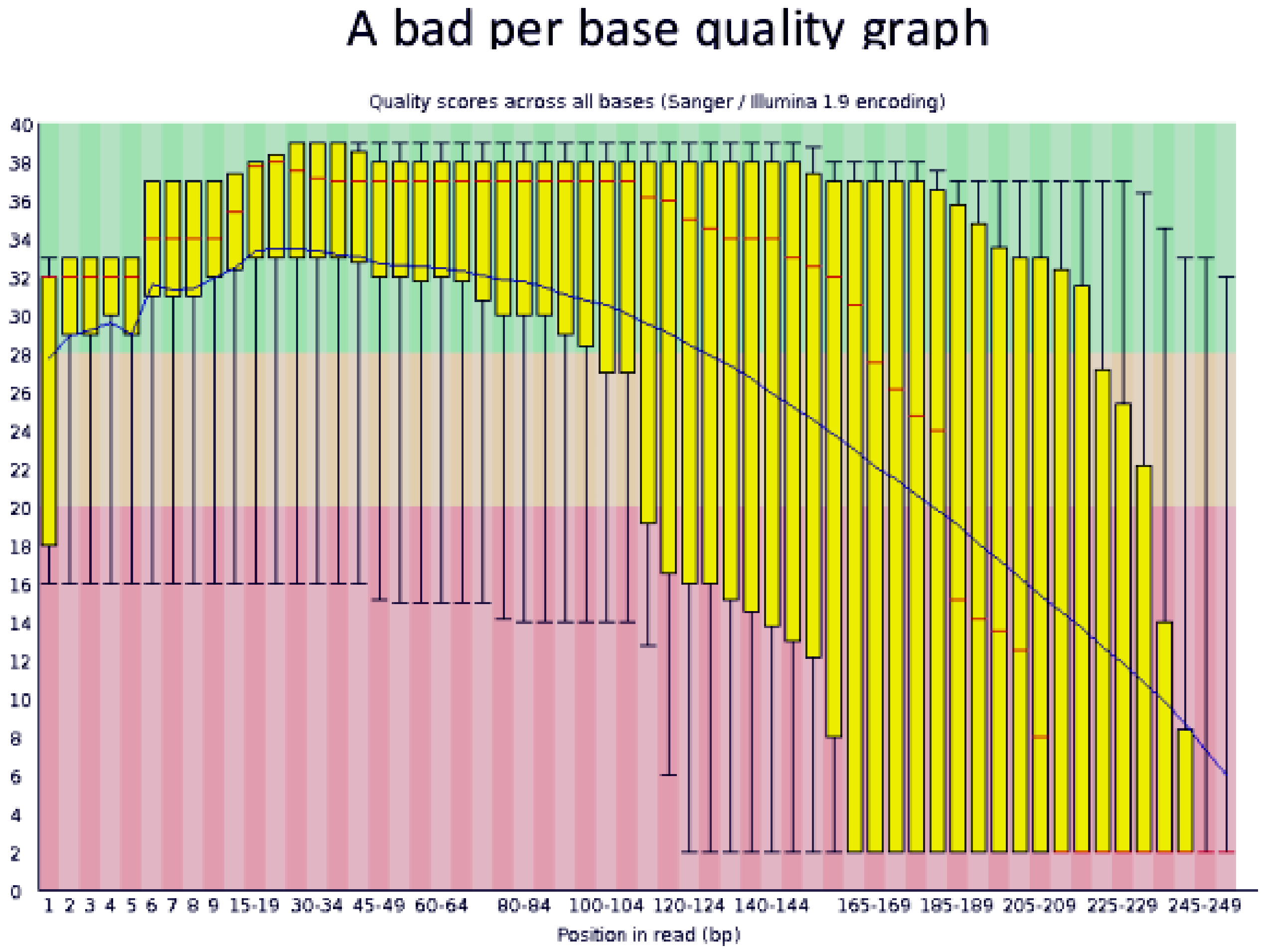
- **Adapter Content** → Foreign Body Detection
  Like detecting a parasite—these shouldn't be in your sample and need removal.



Check Data Quality and Integrity



FastQC – Per base sequence quality



- **Median** (red line)
- **Interquartile range** (yellow box)
- **10th and 90th percentiles** (whiskers)
- **Blue line**: mean quality score.

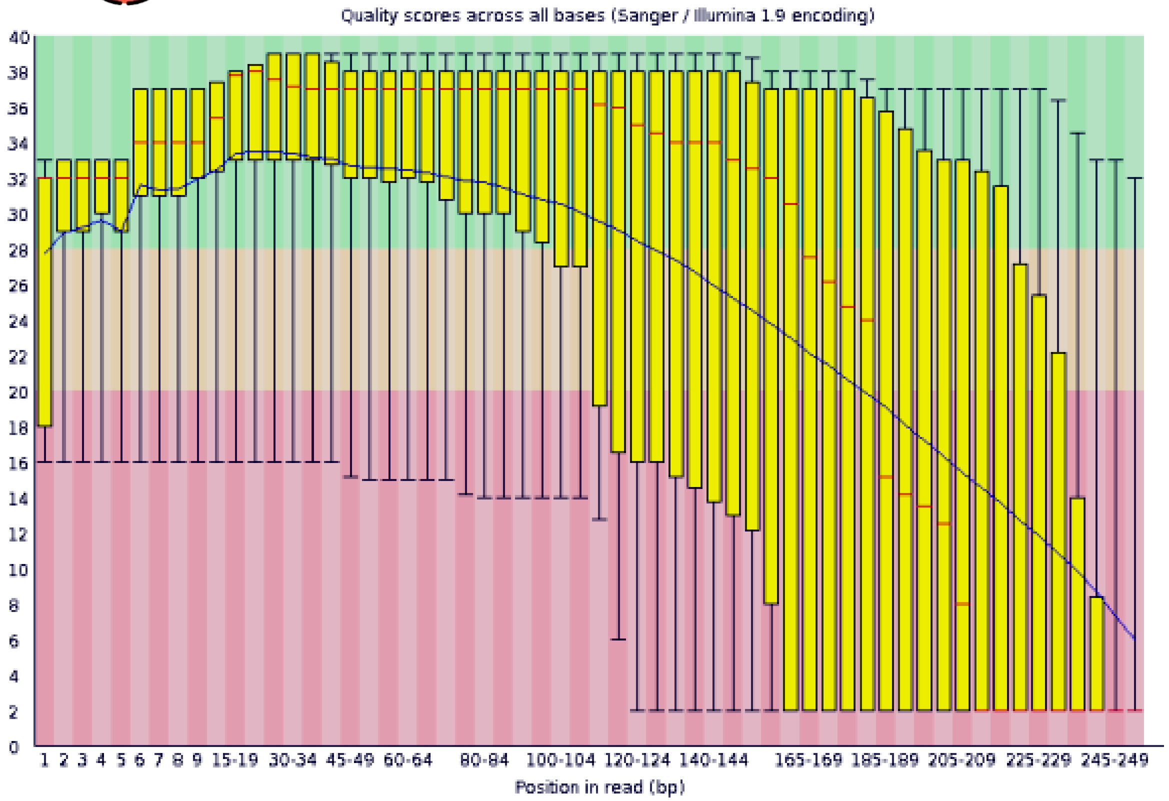
□ **Green** / □ **Orange** / □ **Red Zones**:

👀 **What to watch out for:**

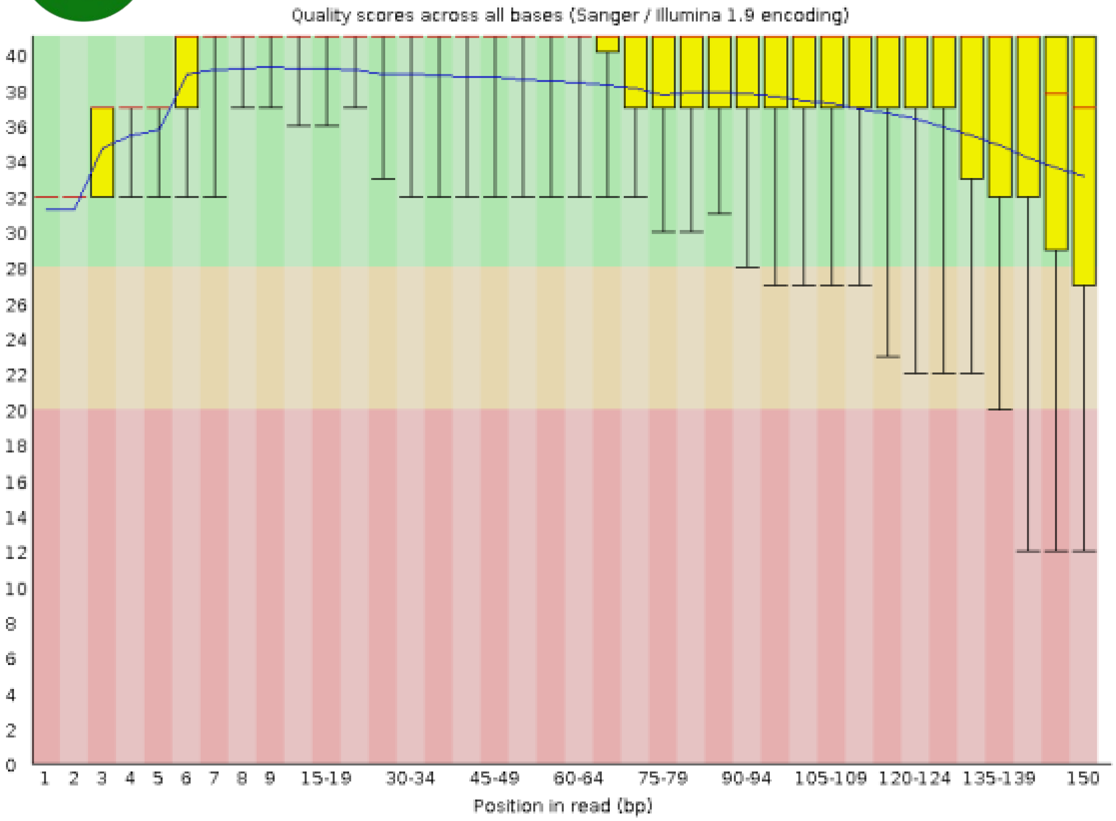
- **Good reads**: High and consistent quality (green zone) across all positions.
- **Drop in quality at the ends**: Common in Illumina data consider trimming.
- **Spikes or dips at specific positions**: May indicate adapter contamination, poor cycles, or technical artifacts.

FastQC – Per base sequence quality

BAD per base quality

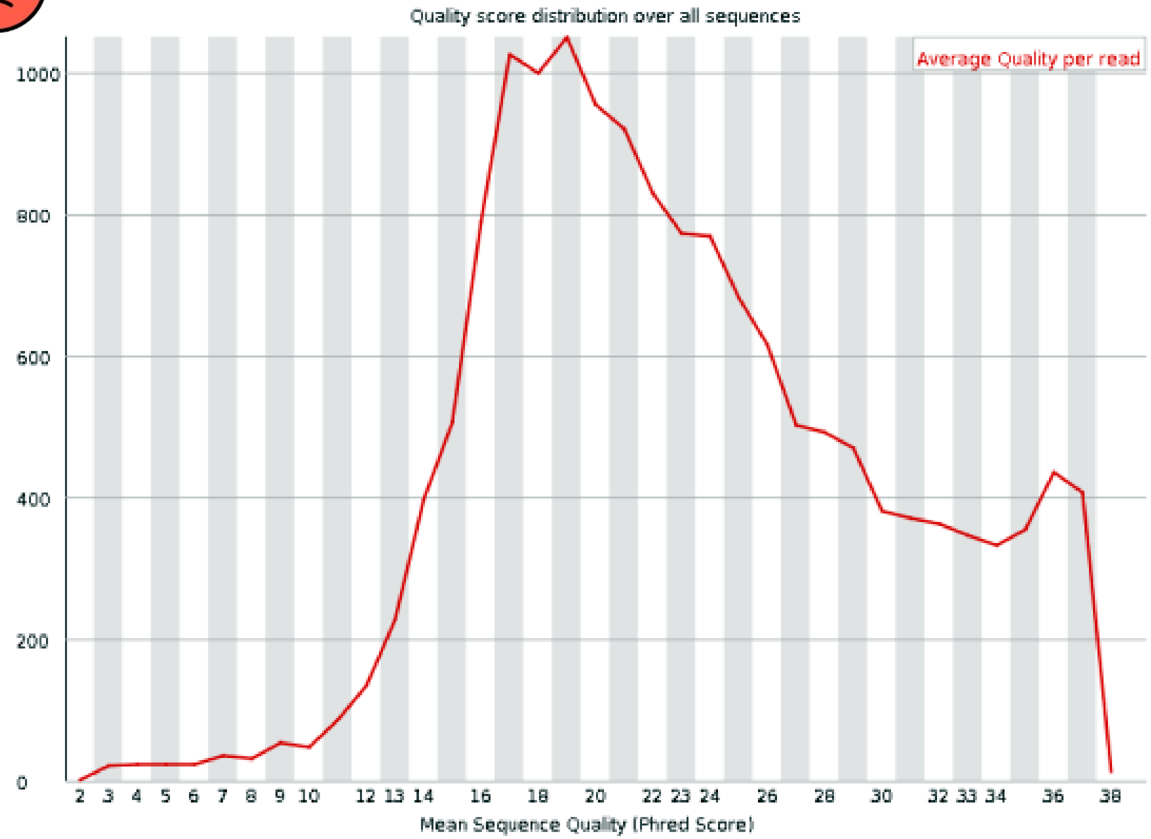


GOOD per base quality



FastQC – Per sequence quality scores

BAD sequence quality graph



- A plot of the total number of reads vs the average quality score over full length of that read.
- **What to look for:** The distribution of average read quality should be fairly tight in the upper range of the plot.

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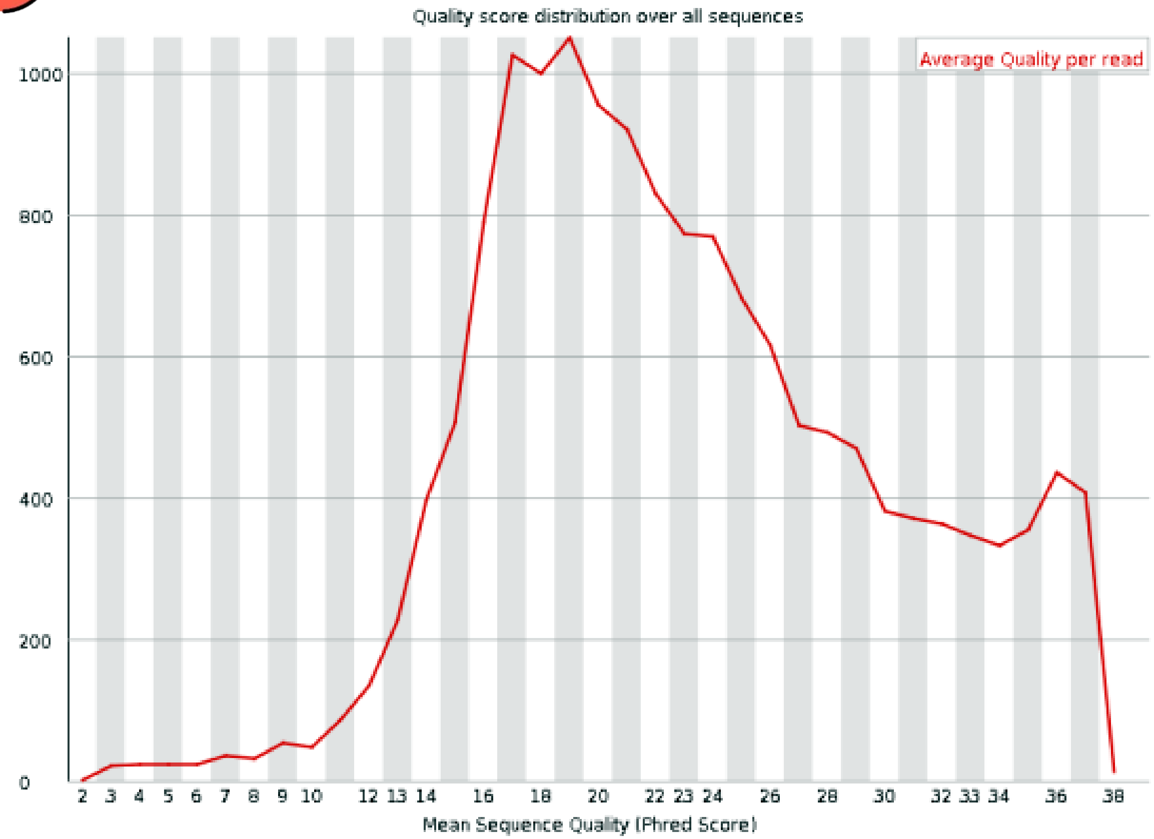
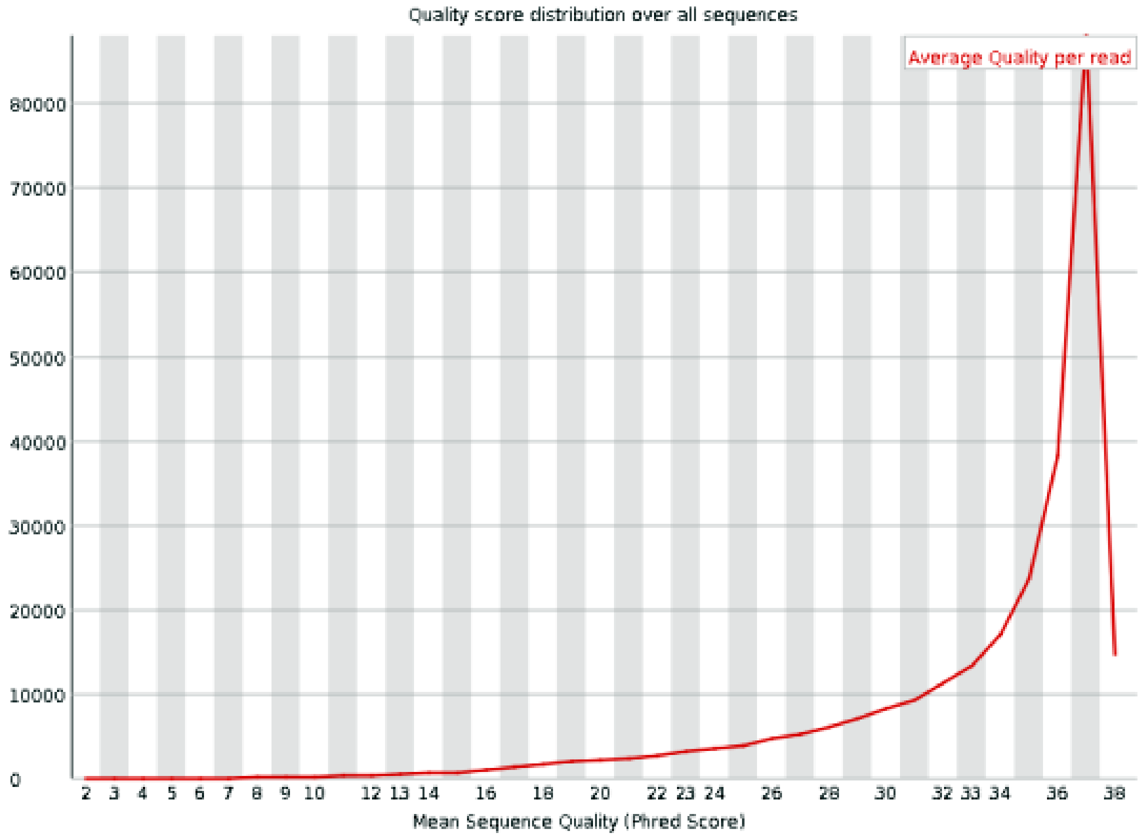
FastQC – Per sequence quality scores

BAD per sequence quality score



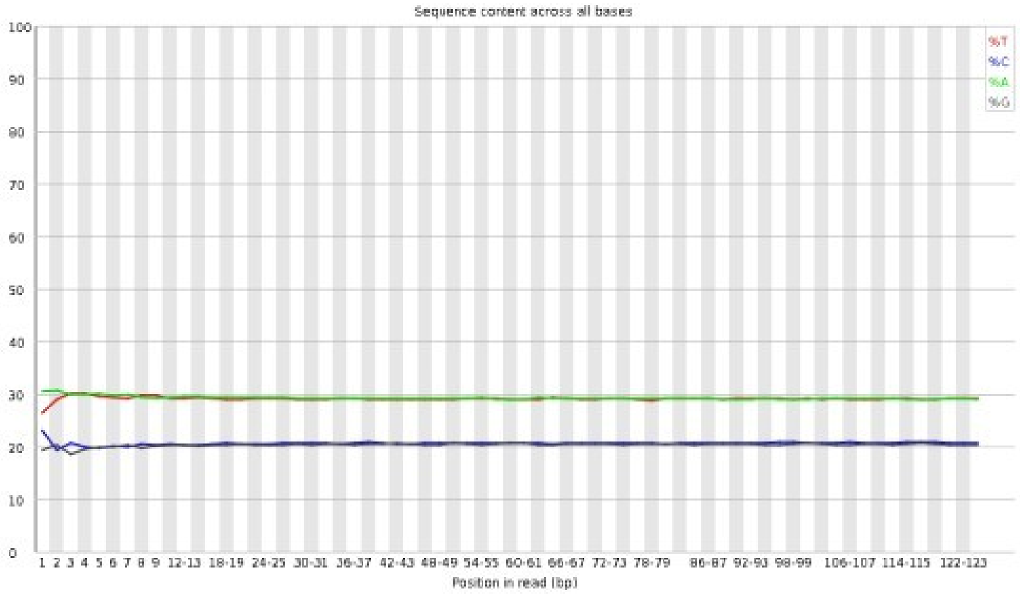
GOOD per sequence quality score



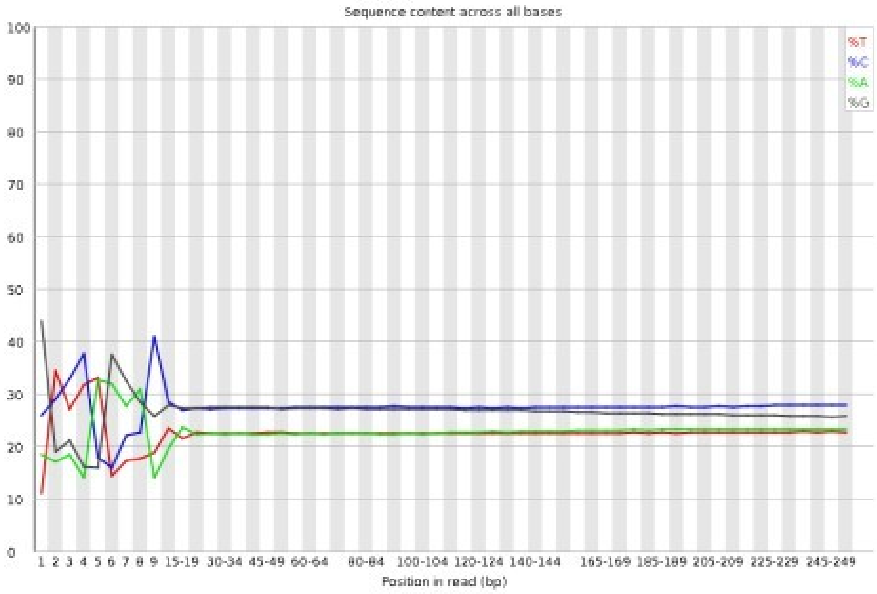
FastQC – Per base sequence content

- A plot reports the percent of bases called for each of the four nucleotides at each position across all reads in the file.
- The proportion of each of the four bases should remain relatively constant over the length of the read with %A=%T and %G=%C.

DNA library per base content



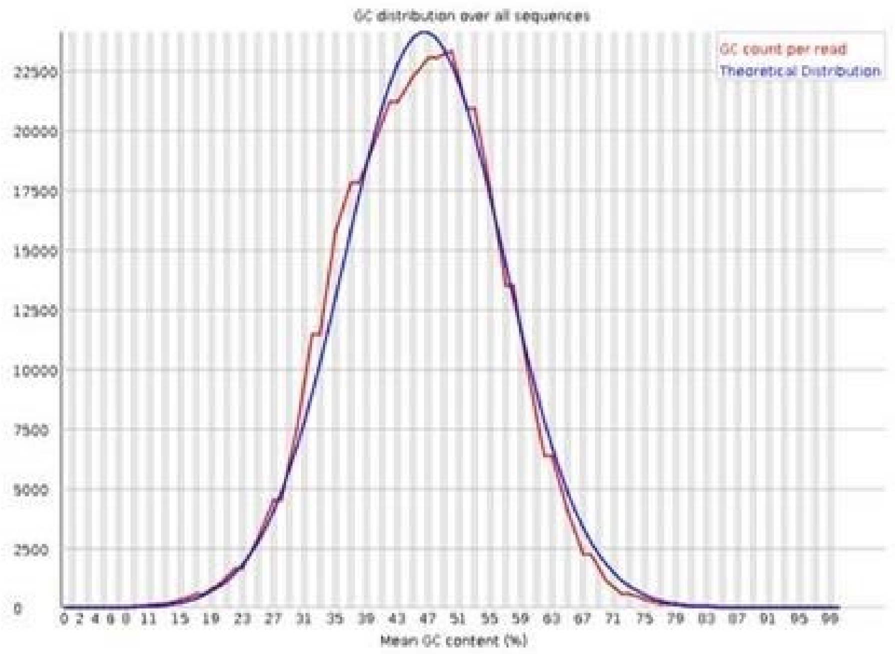
RNA library per base content



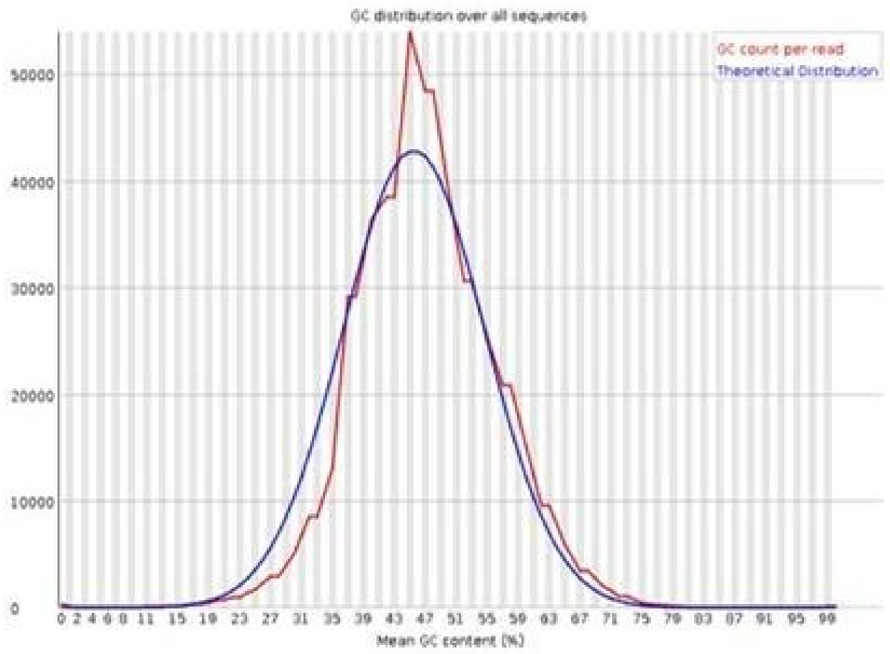
FastQC – Per sequence GC content

What to look for: The expectation is that the GC content of all reads should form a normal distribution with the peak of the curve at the mean GC content for the organism sequenced.

Good Illumina data



Bad Illumina data



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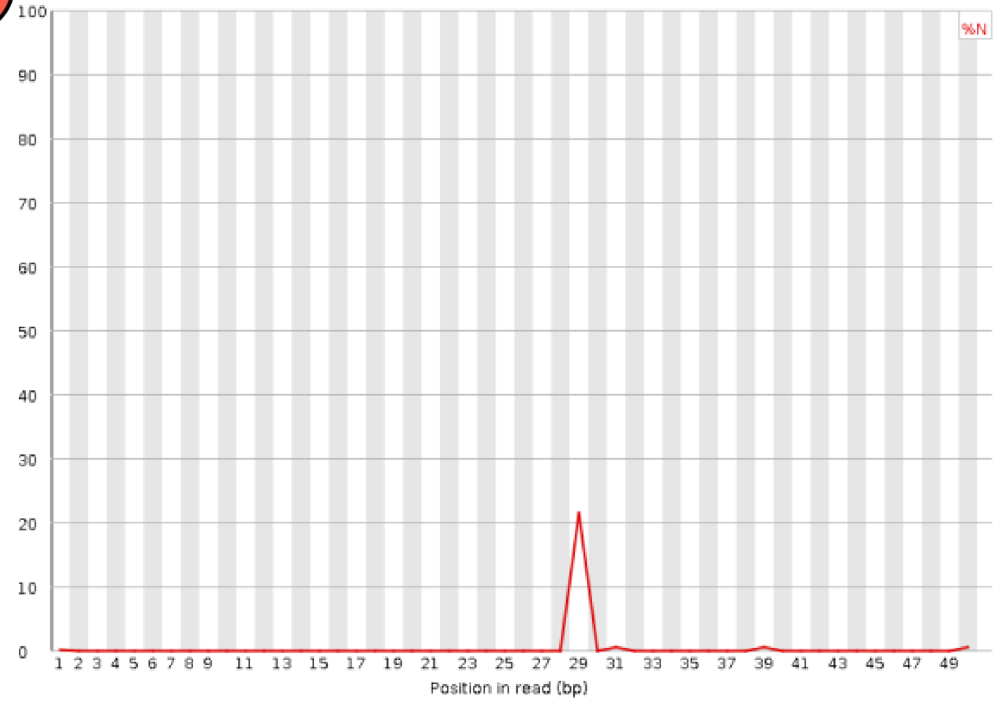
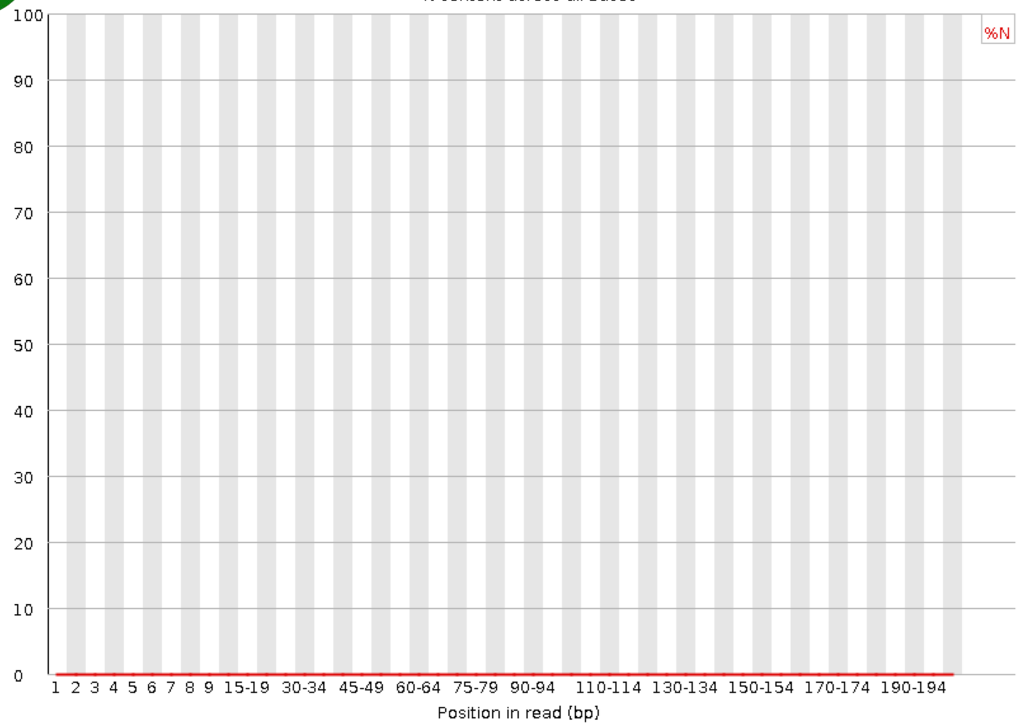
FastQC – Per base N content

- Percent of bases at each position or bin with no base call, i.e. ‘N’.
- **What to expect:** You should never see any point where this curve rises noticeably above zero. If it does this indicates a problem occurred during the sequencing run.

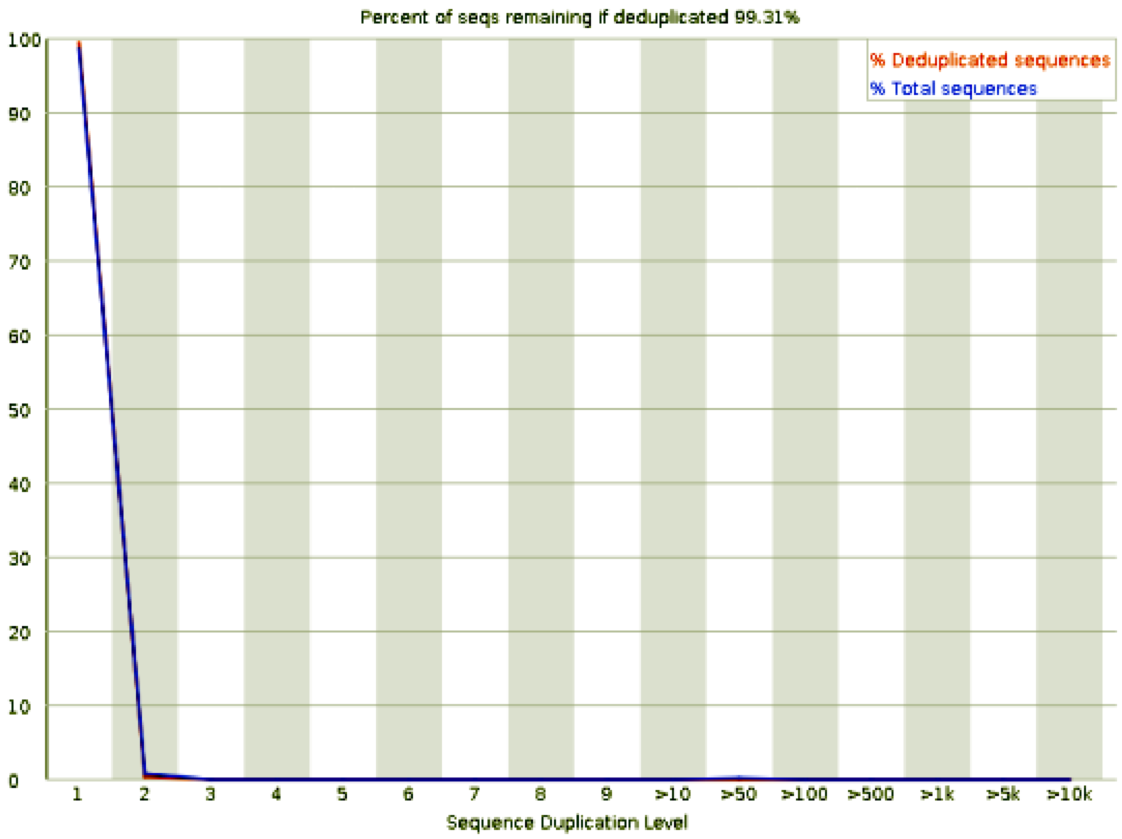
BAD per base N content



GOOD per base N content



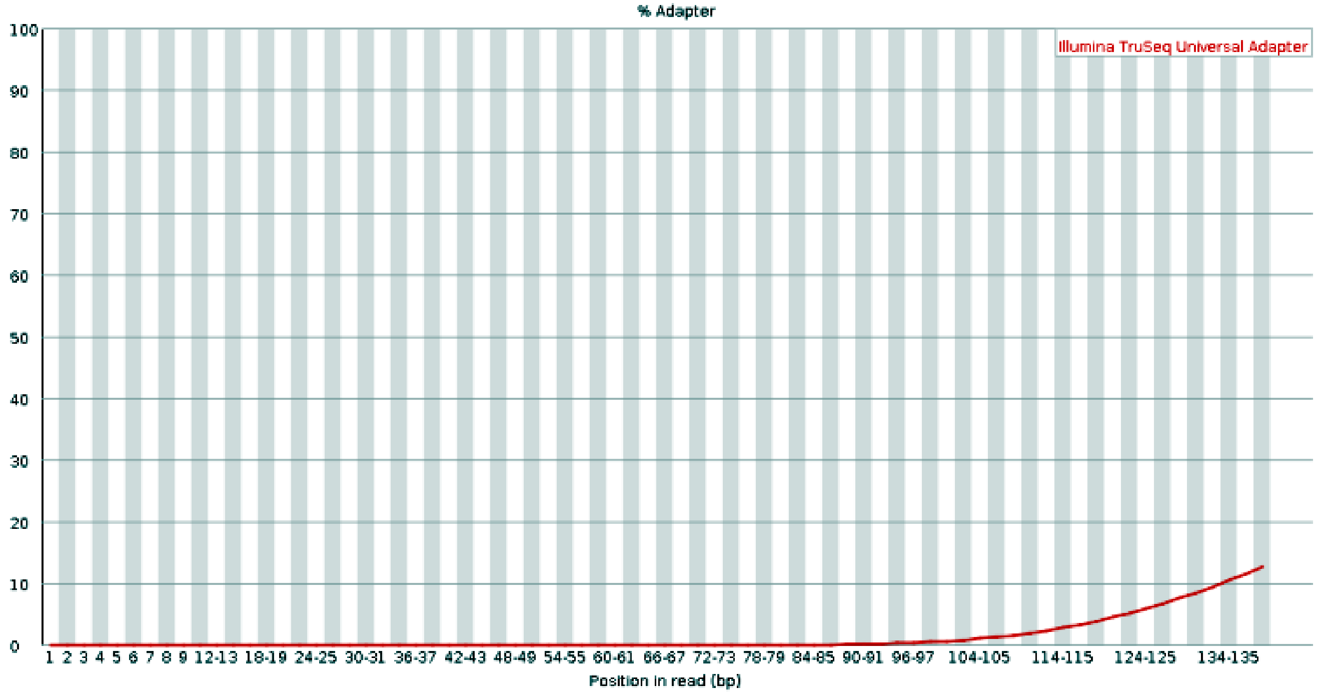
FastQC – Sequence Duplication Levels

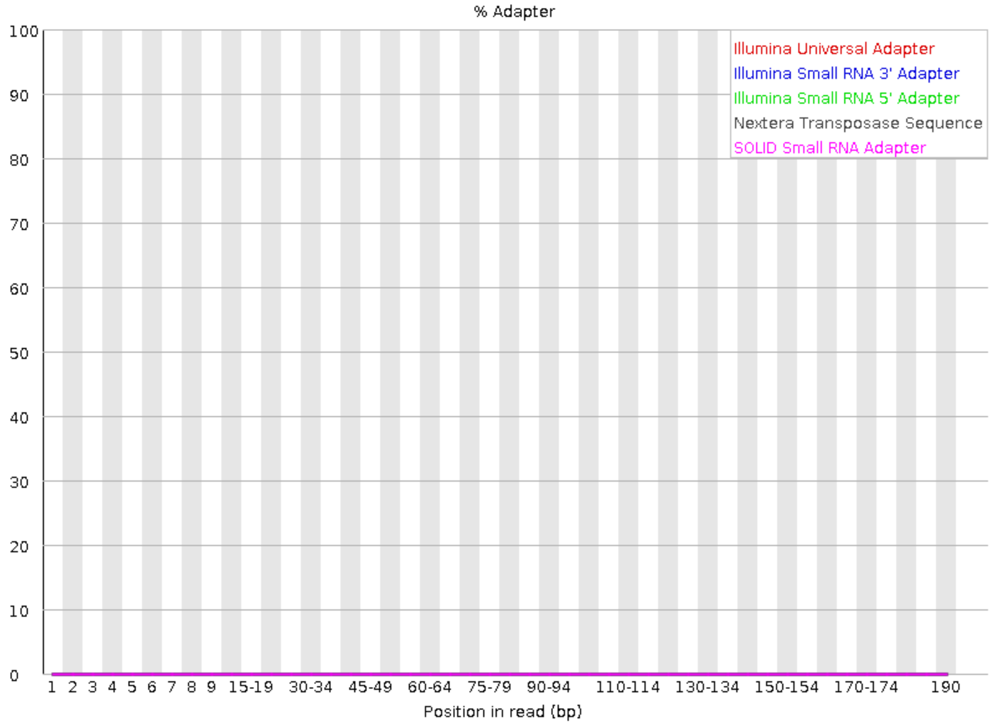


- Percentage of reads of a given sequence in the file which are present a given number of times in the file.
- **Main source: PCR duplication** in which library fragments have been over represented due to biased PCR enrichment.
- **What to expect:** Nearly 100% of your reads should be unique (appearing only 1 time in the sequence data). This indicates a highly diverse library that was not over sequenced.

FastQC – Adapter Content

- Cumulative plot of the fraction of reads where the sequence library adapter sequence is identified at the indicated base position.
- Only adapters specific to the library type are searched.
- **What to expect:** Ideally Illumina sequence data should not have any adapter sequences present.





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MultiQC: Aggregating QC Reports

- MultiQC is a tool that **aggregates results from multiple samples and tools into one report**
- Instead of looking at 50 separate FastQC HTML files, you can run MultiQC on a folder and get a single interactive report summarizing all samples.
- It parses output from dozens of bioinformatics tools (FastQC, fastp, Trimmomatic, Bowtie2, Samtools stats, etc.) and compiles plots/tables for side-by-side comparison.

Using MultiQC:

- After running FastQC on all your FASTQs, simply execute multiqc in the results directory (or specify a path).
- MultiQC will find all supported log/report files automatically
- You can specify output file name with -o or –filename

HOW TO FIX NGS DATA GARBAGE?

Makerere University

IDI Infectious Diseases Institute

African Center of Excellence in Bioinformatics and Data Intensive Sciences

ABI African Bioinformatics Institute

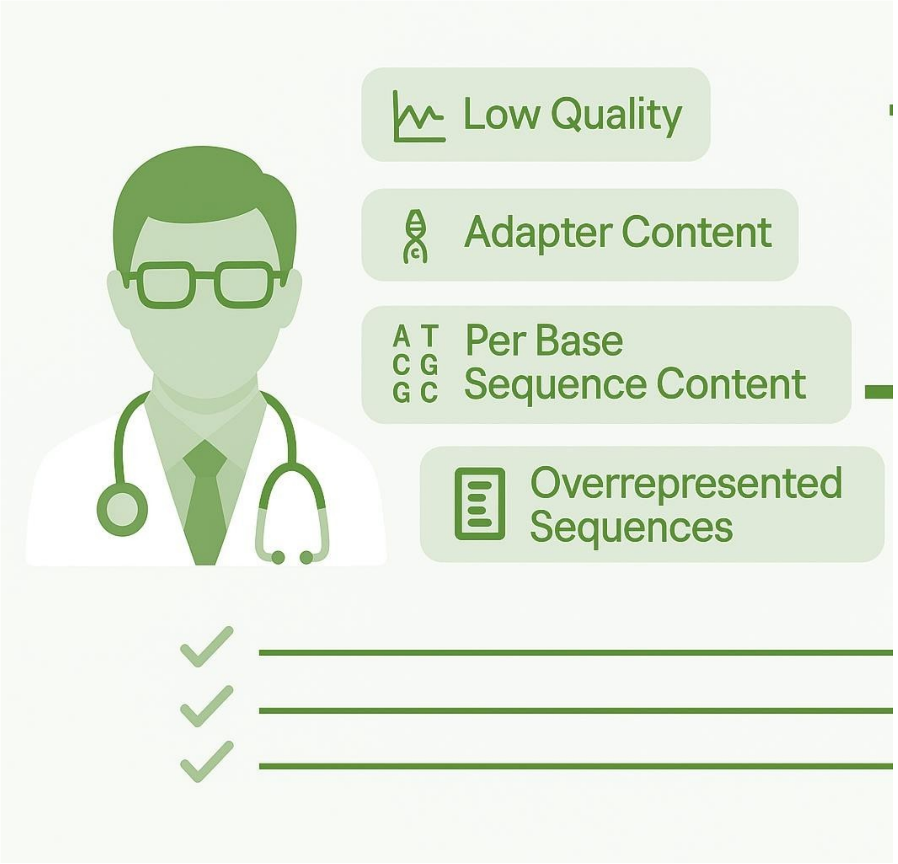
MRC/UVRI and LSHTM Uganda Research Unit

UKRI Medical Research Council

Uganda Virus Research Institute

London School of Hygiene & Tropical Medicine

Read Trimming and Filtering

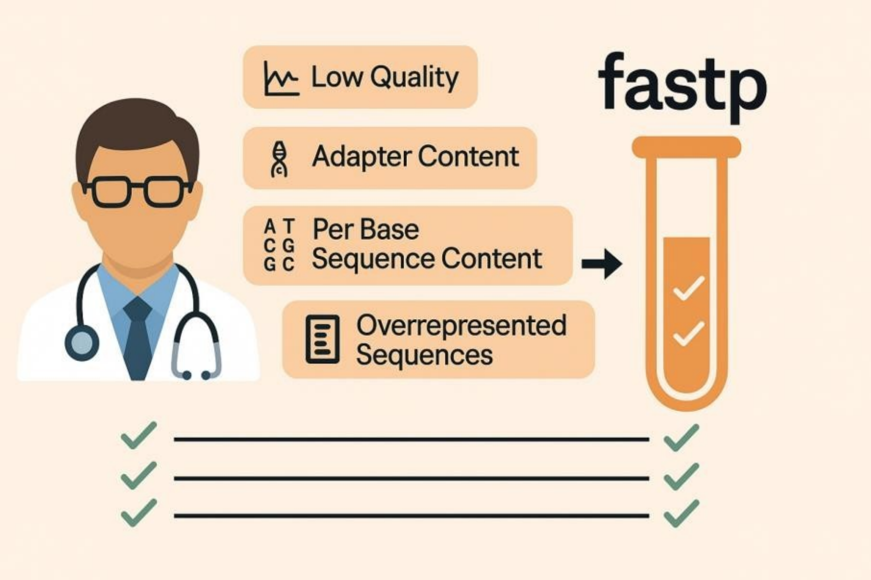


- Removes low-quality bases and adapter sequences.
- Improves the accuracy of read alignment and variant calling.
- Command-line tool with flexible trimming parameters.
- NGS Tools: Trimmomatic, Cutadapt, Fastp, etc
- ONT Tols : Porechop, qcat, Guppy Barcoder, NanoFilt, Filtlong – ONT

fastp: All-in-One QC and Trimming

https://github.com/OpenGene/fastp

```
fastp -i sample_R1.fastq.gz -I sample_R2.fastq.gz \
      -o sample_trim_R1.fastq.gz -O sample_trim_R2.fastq.gz \
      -h sample_fastp.html -j sample_fastp.json
```



- fastp reads sample_R1 and sample_R2, performs default filtering and adapter trimming, and outputs
- It also writes an HTML report (-h) and a JSON summary (-j).
- The default settings will do things like remove reads with >~40% low-quality bases or >5 N's, trim bases below Q15 at ends, etc., which are usually sensible defaults – these can be tuned with options.
- **fastp Reports (pre vs post):** The fastp HTML report is very handy to **compare before/after** QC in one place

PRACTICALS

TASK

Perform comprehensive QC on raw data in rawreads, and discuss the issues identified.